MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 08** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 1: Real Numbers** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

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Q1. If two positive integers a and b are such that $a = 12$ and $\text{HCF}(a, b) = 4$, $\text{LCM}(a, b) = 24$, then b is equal to: [1]

(a) 8 (b) 6 (c) 16 (d) 10

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 8

We know the product identity:

Formula Used: $a \times b = \text{HCF}(a, b) \times \text{LCM}(a, b)$

$12 \times b = 4 \times 24 \implies b = \dfrac{96}{12} = \boxed{8}$. [1M]

Q2. The largest number which divides 70 and 125, leaving remainders 5 and 8 respectively, is: [1]

(a) 65 (b) 13 (c) 17 (d) 5

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 13

Subtracting remainders: $70 - 5 = 65$ and $125 - 8 = 117$. We know:

Formula Used: Required largest number = $\text{HCF}(65, 117)$

Since $65 = 5 \times 13$ and $117 = 3^2 \times 13$, $\text{HCF}(65, 117) = \boxed{13}$. [1M]

Q3. The largest number which divides 318 and 739, leaving remainders 3 and 4 respectively, is: [1]

(a) 105 (b) 15 (c) 35 (d) 210

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 105

Subtracting remainders: $318 - 3 = 315$ and $739 - 4 = 735$. We know:

Formula Used: Required largest divisor = HCF$(315, 735)$

Since $315 = 3^2 \times 5 \times 7 = 3 \times 105$ and $735 = 3 \times 5 \times 7^2 = 7 \times 105$, HCF$(315, 735) = 3 \times 5 \times 7 = \boxed{105}$. [1M]

Q4. The smallest number which when divided by 28 and 32 leaves remainders 8 and 12 respectively is: [1]

(a) 204 (b) 224 (c) 244 (d) 184

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 204

Constant difference: $28 - 8 = 20$ and $32 - 12 = 20$. We know:

Formula Used: Smallest number = LCM$(28, 32) - 20$

With $28 = 2^2 \times 7$ and $32 = 2^5$: LCM$(28, 32) = 2^5 \times 7 = 224$. Subtracting 20: $224 - 20 = \boxed{204}$. [1M]

Q5. Three bells toll together at intervals of $9, 12$, and 15 minutes respectively. If they toll together now, after how many hours will they toll together next? [1]

(a) 2 hours (b) 3 hours (c) 5 hours (d) 6 hours

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) 3 hours

We know simultaneous tolling occurs after LCM$(9, 12, 15)$ minutes:

Formula Used: $9 = 3^2$, $12 = 2^2 \times 3$, $15 = 3 \times 5 \implies$ LCM $= 2^2 \times 3^2 \times 5$

LCM$(9, 12, 15) = 4 \times 9 \times 5 = 180$ minutes $= \dfrac{180}{60} = \boxed{3 \text{ hours}}$. [1M]

Q6. Assertion (A): The HCF of two consecutive natural numbers is 1.
Reason (R): Two consecutive natural numbers are always co-prime. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

Let two consecutive natural numbers be n and $n+1$. Any common divisor d must divide $(n+1) - n = 1 \implies d = 1$. We know:

Formula Used: Co-prime integers $\iff$ HCF $= 1$

Thus consecutive natural numbers are always co-prime (Reason R), which directly implies their HCF is 1 (Assertion A). [1M]

SECTION B: Very Short Answer Type — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. Two numbers are in the ratio 21 : 17. If their HCF is 5, find the numbers. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let the two numbers be $21x$ and $17x$. Since $\text{HCF}(21, 17) = 1$ (co-prime), we know:

Formula Used: $\text{HCF}(21x,\ 17x) = x \cdot \text{HCF}(21, 17) = x$

Given HCF $= 5$, we get $x = 5$. [1M]

Substituting $x = 5$:

$$\text{First number} = 21 \times 5 = \boxed{105}, \qquad \text{Second number} = 17 \times 5 = \boxed{85}. \qquad [1M]$$

Q8. Find the largest number that divides 2053 and 367, leaving remainders 5 and 7 respectively. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Subtracting the given remainders from 2053 and 367:

$$2053 - 5 = 2048, \qquad 367 - 7 = 360. \qquad [1M]$$

We know the largest such divisor is:

Formula Used: Required divisor $= \text{HCF}(2048, 360)$

Factorising: $2048 = 2^{11}$ and $360 = 2^3 \times 3^2 \times 5$ (or by Euclid: $2048 = 360 \times 5 + 248$, $360 = 248 \times 1 + 112$, $248 = 112 \times 2 + 24$, $112 = 24 \times 4 + 16$, $24 = 16 \times 1 + 8$, $16 = 8 \times 2 + 0$). Common prime factor is $2^3 = \boxed{8}$. [1M]

SECTION C: Short Answer Type — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Find the HCF and LCM of $144, 180$, and 192 by prime factorisation, and show that HCF $\times$ LCM $\neq$ Product of the three numbers. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Prime factorising $144, 180$, and 192:

$$144 = 2^4 \times 3^2, \qquad 180 = 2^2 \times 3^2 \times 5, \qquad 192 = 2^6 \times 3. \qquad [1M]$$

We know for three integers a, b, c:

Formula Used: $\text{HCF} = \prod p^{\min \text{ exp}}$, $\text{LCM} = \prod p^{\max \text{ exp}}$, generally $\text{HCF} \cdot \text{LCM} \neq abc$

Computing HCF and LCM:

$$\text{HCF} = 2^2 \times 3^1 = \boxed{12}, \qquad \text{LCM} = 2^6 \times 3^2 \times 5 = 64 \times 45 = \boxed{2880}. \qquad [1M]$$

Comparing HCF $\times$ LCM with $144 \times 180 \times 192$:

$$\text{HCF} \times \text{LCM} = 12 \times 2880 = \boxed{34560}, \qquad 144 \times 180 \times 192 = 25920 \times 192 = \boxed{4976640}.$$

Clearly $34560 \neq 4976640 \implies$ HCF $\times$ LCM $\neq$ Product of the three numbers. **[1M]**

Q10. If $d = \text{HCF}(48, 72)$, find integers x and y such that $d = 48x + 72y$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

First finding $d = \text{HCF}(48, 72)$ via prime factorisation or Euclid's division step:

$$48 = 2^4 \times 3, \qquad 72 = 2^3 \times 3^2 \implies d = \text{HCF}(48, 72) = 2^3 \times 3 = \boxed{24}. \qquad [1M]$$

We apply Euclid's division relation (Bézout's linear combination):

Formula Used: $72 = 48 \times 1 + 24 \implies 24 = 72 - 48 = 48(-1) + 72(1)$

Comparing $24 = 48(-1)+72(1)$ with $d = 48x+72y$, one immediate integer solution is $\boxed{x = -1,\ y = 1}$. **[1M]**

Note (Non-uniqueness): Dividing $48x + 72y = 24$ by 24 gives $2x + 3y = 1$, which yields infinitely many integer pairs $(x, y) = (-1 + 3k,\ 1 - 2k)$ for $k \in \mathbb{Z}$ (e.g., for $k = 1$: $\boxed{x = 2,\ y = -1}$ since $48(2) + 72(-1) = 96 - 72 = 24$). **[1M]**

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Q11. Prove that $\sqrt{5}$ is an irrational number. Hence, prove that $(3\sqrt{5}-2)^2$ is an irrational number. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Part 1 (3 Marks) – Proving $\sqrt{5}$ is irrational:

Suppose, on the contrary, that $\sqrt{5} = \frac{a}{b}$ where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\text{HCF}(a, b) = 1$.

Squaring both sides: $a^2 = 5b^2$ — (1) [1M]

We know the Fundamental Theorem of Arithmetic prime divisibility lemma:

Formula Used: If a prime p divides a^2, then p divides a.

Since 5 divides a^2, 5 divides a, so $a = 5c$ for some integer c. Substituting in (1):

$$(5c)^2 = 5b^2 \implies b^2 = 5c^2 \implies 5 \text{ divides } b^2 \implies 5 \text{ divides } b. \quad [1M]$$

Thus 5 is a common factor of a and b, contradicting $\text{HCF}(a, b) = 1$. Hence $\boxed{\sqrt{5} \text{ is irrational.}}$ [1M]

Part 2 (2 Marks) – Proving $(3\sqrt{5}-2)^2$ is irrational:

Expanding $(3\sqrt{5}-2)^2$ using $(a-b)^2 = a^2 + b^2 - 2ab$:

$$(3\sqrt{5}-2)^2 = (3\sqrt{5})^2 + 2^2 - 2(3\sqrt{5})(2) = 45 + 4 - 12\sqrt{5} = 49 - 12\sqrt{5}. \quad [1M]$$

Suppose, to the contrary, $49 - 12\sqrt{5} = \frac{p}{q}$ $(p, q \in \mathbb{Z},\ q \neq 0)$. Rearranging for $\sqrt{5}$:

$$12\sqrt{5} = 49 - \frac{p}{q} = \frac{49q - p}{q} \implies \sqrt{5} = \frac{49q - p}{12q}.$$

Because $p, q \in \mathbb{Z}$ and $12q \neq 0$, $\frac{49q-p}{12q} \in \mathbb{Q}$, contradicting Part 1 ($\sqrt{5} \notin \mathbb{Q}$). Thus $\boxed{(3\sqrt{5}-2)^2 \text{ is irrational.}}$ [1M]

Q12. Prove that if p is a prime number, then $\sqrt{p}$ is irrational. Hence, prove that $\sqrt{2}+\sqrt{5}$ is an irrational number. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Part 1 (3 Marks) – Proving $\sqrt{p}$ is irrational (where p is a prime):

Suppose, on the contrary, that $\sqrt{p}=\frac{a}{b}$ where $a,b\in\mathbb{Z}$, $b\neq 0$, and $\text{HCF}(a,b)=1$.

Squaring both sides: $a^2=pb^2$ — (1) **[1M]**

We know the Fundamental Theorem of Arithmetic prime divisibility lemma:

Formula Used: If a prime p divides a^2, then p divides a.

Since p divides a^2, p divides a, so $a=pc$ for some integer c. Substituting in (1):

$$(pc)^2=pb^2 \implies b^2=pc^2 \implies p \text{ divides } b^2 \implies p \text{ divides } b. \quad \textbf{[1M]}$$

Thus p is a common factor of a and b, contradicting $\text{HCF}(a,b)=1$. Hence $\boxed{\sqrt{p} \text{ is irrational.}}$ **[1M]**

Part 2 (2 Marks) – Proving $\sqrt{2}+\sqrt{5}$ is irrational:

Since 2 and 5 are prime, $\sqrt{2}$ and $\sqrt{5}$ are irrational by Part 1. Suppose $r=\sqrt{2}+\sqrt{5}\in\mathbb{Q}$ $(r>0)$. Writing $\sqrt{5}=r-\sqrt{2}$ and squaring both sides:

$$5=r^2+2-2r\sqrt{2} \implies 2r\sqrt{2}=r^2-3 \implies \sqrt{2}=\frac{r^2-3}{2r}. \quad \textbf{[1M]}$$

Since $r\in\mathbb{Q}\setminus\{0\}$, $\frac{r^2-3}{2r}\in\mathbb{Q}$, contradicting the irrationality of $\sqrt{2}$ proved in Part 1. Hence $\boxed{\sqrt{2}+\sqrt{5} \text{ is irrational.}}$ **[1M]**

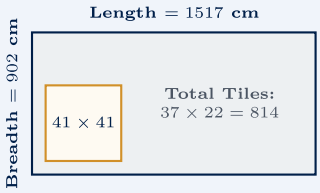
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Q13. CASE STUDY 08: HERITAGE PALACE ARCHITECTURAL CERAMIC TILE PAVING [4 Marks]

An architectural restoration team is renovating a magnificent ceremonial banquet courtyard in a heritage palace. The rectangular courtyard measures 15 m 17 cm in length and 9 m 2 cm in breadth. The chief architect specifies that the entire courtyard must be paved with the minimum possible number of identical square ceramic tiles, cut without any material wastage.

(i) Determine the largest side length of the square tile in centimetres. [2]
(ii) Find the minimum number of tiles required to pave the entire courtyard. [1]
(iii) Calculate the perimeter of one single square tile in centimetres. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Geometric Paving: To pave a 1517 cm × 902 cm rectangle without cutting, the square side s must divide both dimensions; minimizing tile count chooses $s = \text{HCF}(1517, 902) = 41$ cm.

Converting dimensions to centimetres: $L = 15$ m 17 cm $= 1517$ cm, $B = 9$ m 2 cm $= 902$ cm. We know:

Formula Used: Max tile side $s = \text{HCF}(L, B)$, Min tiles $= \dfrac{L \times B}{s^2}$, Tile perimeter $= 4s$

(i) Largest tile side length: Prime factorising 1517 and 902:

$$1517 = 37 \times 41, \qquad 902 = 2 \times 11 \times 41 \implies s = \text{HCF}(1517, 902) = \boxed{41 \text{ cm}}. \qquad [2M]$$

(ii) Minimum number of tiles: Dividing length and breadth by 41 cm:

$$\text{Total tiles} = \frac{1517}{41} \times \frac{902}{41} = 37 \times 22 = \boxed{814 \text{ tiles}}. \qquad [1M]$$

(iii) Perimeter of one square tile: $P = 4 \times s = 4 \times 41 = \boxed{164 \text{ cm}}$. [1M]

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